

HUJI Applicative Research Report — Week 38, 2026

5 high-potential paper(s) · Weekly · Week 38, 2026 · Generated September 14, 2026

The highlighted research showcases pioneering advancements across diverse therapeutic and technological fronts. Breakthroughs in regenerative medicine include an off-the-shelf stem cell therapy for demyelinating disorders and a biologic for joint repair. Additionally, a novel gene therapy platform addresses cellular delivery challenges, while another targets a severe pediatric epilepsy. Finally, a new metasurface technology promises transformative applications in stealth and thermal management.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Tamir Ben-Hur** — Neuroscience, Immunology, Clinical, Synthetic Biology
- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Amir Haze** — Medical Device, Clinical, Synthetic Biology, Proteomics
- **Rami I Aqeilan** — Clinical, Neuroscience, Genomics
- **Roy Maman** — Materials, Imaging

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders. 36/50

"Off-the-shelf stem cell therapy for demyelinating disorders offers new hope for MS patients."

Main researcher: Tamir Ben-Hur · tamir@hadassah.org.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; The Department of Neurology, The Agnes-Ginges Center for Human Neurogenetics, Hadassah-Hebrew University Medical Center, Jerusalem, Israel. Electronic address: tamir@hadassah.org.il.

Published in Stem cell reports · 2026-06

ABOUT THIS RESEARCH

Researchers developed a clinical-grade astrocyte therapy derived from embryonic stem cells that helps the brain repair damaged myelin sheaths. The therapy works by clearing debris and creating a supportive environment that encourages the growth of new insulating layers for nerves.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

COMMERCIALISATION METRICS

Novelty

6/10

While the use of hESC-derived astrocytes is a sophisticated approach to remyelination, the 'bystander mechanism' and the focus on microglial debris clearance are established concepts in neuroregeneration, making this more of an incremental application of cell therapy than a groundbreaking scientific paradigm shift.

Commercial Potential

8/10

The therapy uses clinical-grade, scalable hESC-derived cells targeting a high-unmet-need market (demyelinating disorders) with demonstrated safety and a clear mechanism of action. The 'clinical-grade' status and scalability suggest it is already positioned for translational development rather than being purely theoretical.

Market Size

9/10

The therapy targets demyelinating disorders (such as Multiple Sclerosis), which represent a massive, high-value patient population with significant unmet medical needs. The mechanism's ability to address multiple facets of remyelination failure suggests a broad, multi-indication platform potential within the neurodegenerative market.

Tech Readiness

5/10

The technology has transitioned from basic lab discovery to 'clinical-grade' manufacturing and has demonstrated efficacy in animal models (lysolecithin-induced), but it remains in the preclinical validation stage rather than undergoing human clinical trials.

IP Strength

8/10

The development of a 'clinical-grade' hESC-derived astrocyte product (AstroRx) suggests a proprietary, scalable manufacturing process and a specific cell composition that is highly patentable. The multi-modal mechanism of action—addressing debris clearance and OPC differentiation—provides multiple layers of therapeutic defensibility and platform potential for various CNS disorders.

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2. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake.

33/50

"Breakthrough in gene delivery: a new platform for stable, intracellular genetic therapies."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

ABOUT THIS RESEARCH

Researchers developed a new type of synthetic DNA analogue called MCP PNA that can naturally enter cells without needing complex delivery vehicles. This solves a major historical bottleneck where these stable molecules were too large or poorly charged to cross cell membranes.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty

6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential

8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size

9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness

3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength

7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

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3. Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model. 33/50

"Regenerative protein biologic shows strong potential for healing joint injuries and osteoarthritis."

Main researcher: Amir Haze

Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.; Orthopedic Surgery Department, Hadassah-Hebrew University Medical Center, Jerusalem, Israel.; The Wohl Institute for Translational Medicine, Hadassah-Hebrew University Medical Center, Jerusalem, Israel.

Published in Journal of orthopaedic research : official publication of the Orthopaedic Research Society · 2026-04

ABOUT THIS RESEARCH

Researchers tested a recombinant human amelogenin protein product called Remelix® in a goat model to treat cartilage and bone injuries. The study showed that the protein significantly promotes the healing of both cartilage and subchondral bone without adverse effects.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

COMMERCIALISATION METRICS

Novelty

4/10

The research focuses on validating a known biological mechanism (amelogenin) in a larger animal model rather than introducing a new molecular entity or novel platform. While the transition from rat to goat is a critical translational step, it represents incremental refinement of existing prior art rather than a groundbreaking scientific innovation.

Commercial Potential

8/10

The transition from rodent models to a large animal (goat) model significantly de-risks the technology and demonstrates clear readiness for clinical translation. The mention of a specific, branded product (Remelix®) and a defined carrier (PX 407) suggests a high degree of developmental maturity and a clear path toward a regulated medical device or biologic.

Market Size

8/10

The target market for osteochondral repair is substantial, driven by a high prevalence of joint injuries and the significant clinical unmet need for regenerative solutions in an aging population. The transition from rat to large animal models suggests a high potential for clinical translation into a large, high-value orthopedic market.

Tech Readiness

6/10

The study has progressed from small animal models to a large animal (goat) model, which is a critical regulatory milestone for orthopedic biologics. However, it remains in the preclinical stage and has not yet entered human clinical trials.

IP Strength

7/10

The use of a branded recombinant protein (Remelix®) suggests established manufacturing processes and likely composition-of-matter or process patents. However, the reliance on a well-known protein like amelogenin may face challenges regarding non-obviousness compared to existing recombinant growth factor technologies.

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4. Neuron-specific WWOX gene therapy produces dose-dependent, durable rescue in a model of WWOX-related epileptic encephalopathy.

32/50

"Neuron-specific gene therapy delivers durable rescue for severe pediatric epilepsy."

Main researcher: Rami I Aqeilan

The Concern Foundation Laboratories, The Lautenberg Center for Immunology and Cancer Research, Department of Immunology and Cancer Research-IMRIC, Hebrew University-Hadassah Medical School, Jerusalem, Israel.

Published in Molecular therapy. Advances · 2026 Sep 10

ABOUT THIS RESEARCH

Hebrew University researchers developed a neuron-specific gene therapy that achieves long-term neurological improvement in a mouse model of WWOX-related epileptic encephalopathy. The treatment demonstrates a dose-dependent response, suggesting a viable therapeutic pathway for addressing this severe pediatric neurodevelopmental disorder.

COMMERCIAL ANGLE

This research provides a foundational asset for a clinical-stage orphan drug program targeting pediatric epilepsy through adeno-associated virus (AAV) mediated gene replacement therapy.

COMMERCIALISATION METRICS

Novelty

7/10

The research represents a significant translational advance in gene therapy for a rare neurodevelopmental disorder by achieving durable, neuron-specific rescue; however, it builds upon established AAV-mediated CNS delivery paradigms rather than introducing a radically new therapeutic modality.

Commercial Potential

8/10

Gene therapy for rare, monogenic neurological disorders represents a high-value, clinically validated modality with strong potential for orphan drug designation and rapid regulatory pathways. The clear mechanistic focus on a specific gene target facilitates a defined development roadmap and high likelihood of interest for biotech acquisition or licensing.

Market Size

3/10

WWOX-related epileptic encephalopathy is an ultra-rare orphan disease with a very small patient population, limiting the total addressable market size despite the high unmet medical need. While gene therapy provides high value per patient, the narrow target indication restricts the potential for significant commercial scale compared to broader neurological disorders.

Tech Readiness

6/10

The study demonstrates proof-of-concept efficacy in an animal model with a viral delivery system, indicating a late preclinical stage of development. While highly promising for a specific rare disease, it has yet to initiate human clinical trials or establish large-scale manufacturing processes required for commercialization.

IP Strength

8/10

Gene therapy constructs targeting specific loci (WWOX) combined with tissue-specific promoters represent highly protectable compositions of matter with clear regulatory exclusivity pathways. The approach enables a distinct therapeutic platform for neuro-genetic disorders, providing a strong foundation for a robust, multi-layered patent estate.

[View in Dashboard Open Paper](#)

5. Metasurface platform for simultaneous and uncorrelated emissivity control over MWIR and LWIR spectral bands.32/50

"Metasurface technology revolutionizes thermal control for stealth and advanced electronics."

Main researcher: Roy Maman · Roy.maman@mail.huji.ac.il.

Institute of Applied Physics, The Faculty of Science, The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem, 91904, Israel.

Published in Light, science & applications · 2026-07

ABOUT THIS RESEARCH

Researchers have developed a metasurface capable of independently controlling how an object emits heat (thermal radiation) across two different infrared spectrums. This allows for the precise manipulation of thermal signatures and heat dissipation without one wavelength affecting the other.

COMMERCIAL ANGLE

High-value applications in stealth technology, advanced thermal management for electronics, and specialized non-invasive thermal imaging sensors.

COMMERCIALISATION METRICS

Novelty7/10

The ability to achieve uncorrelated, simultaneous control over two distinct infrared bands (MWIR and LWIR) represents a significant functional leap over single-band or correlated metasurfaces, though it builds on existing plasmonic/dielectric resonator principles.

Commercial Potential7/10

The ability to independently control MWIR and LWIR emissivity creates a high-value platform for thermal camouflage, energy-efficient coatings, and advanced sensing. While the technical application is clear, the score is tempered by the lack of provided data on scalability and manufacturing costs.

Market Size7/10

The ability to independently control MWIR and LWIR emissivity is a platform-enabling capability with high value in defense (stealth/cloaking) and industrial thermal management, though it targets specialized high-end niches rather than a mass consumer market.

Tech Readiness3/10

The paper describes the development of a metasurface platform for spectral control, which is characteristic of fundamental materials science and lab-stage proof-of-concept work. There is no evidence of integration into a commercial device, field testing, or scalable manufacturing processes.

IP Strength8/10

The development of a platform for uncorrelated spectral control in two distinct bands represents a high-value 'platform-enabling' innovation with strong potential for broad patent claims across defense and industrial sensing.

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